

PIZZA SALES ANALYSIS

1. Project Overview:

This project analyzes pizza sales using transactional data from 21,350 orders across various product categories. The goal is analyze key indicators for our pizza sales data to gain insights into our business performance, visualize various aspects of our pizza sales data to gain insights and understand key trends.

2. Dataset Summary:

- Rows: 48,621

- Columns: 13

- Key Features:

- + Customer orders (pizza_id, order_id, pizza_name_id, quantity, order_date, order_day, order_time.

- + Purchase details (unit_price, total_price, pizza_size, pizza_category, pizza_ingredients, pizza_name)

- Missing Data: No

3. Exploratory Data Analysis using Excel:

We began with data preparation and cleaning in Excel:

- **Data Loading:** Create a copy of CSV data file and use Ctr+T for create table

pizza_id	order_id	total_orders	pizza_name_id	quantity	order_date	order_day	order_time	unit_price	total_price	pizza_size	pizza_category
1	1	1	hawaiian_m	1	01/01/2015	Thursday	11:38:36 AM	13,25	13,25	M	Classic
2	2	0,2	classic_dlx_m	1	01/01/2015	Thursday	11:57:40 AM	16	16	M	Classic
3	2	0,2	five_cheese_l	1	01/01/2015	Thursday	11:57:40 AM	18,5	18,5	L	Veggie
4	2	0,2	ital_supr_l	1	01/01/2015	Thursday	11:57:40 AM	20,75	20,75	L	Supreme
5	2	0,2	mexicana_m	1	01/01/2015	Thursday	11:57:40 AM	16	16	M	Veggie
6	2	0,2	thai_ckn_l	1	01/01/2015	Thursday	11:57:40 AM	20,75	20,75	L	Chicken
7	3	0,5	ital_supr_m	1	01/01/2015	Thursday	12:12:28 PM	16,5	16,5	M	Supreme
8	3	0,5	prsc_argla_l	1	01/01/2015	Thursday	12:12:28 PM	20,75	20,75	L	Supreme
9	4	1	ital_supr_m	1	01/01/2015	Thursday	12:16:31 PM	16,5	16,5	M	Supreme
10	5	1	ital_supr_m	1	01/01/2015	Thursday	12:21:30 PM	16,5	16,5	M	Supreme
11	6	0,5	bbq_ckn_s	1	01/01/2015	Thursday	12:29:36 PM	12,75	12,75	S	Chicken
12	6	0,5	the_greek_s	1	01/01/2015	Thursday	12:29:36 PM	12	12	S	Classic
13	7	1	spinach_supr_s	1	01/01/2015	Thursday	12:50:37 PM	12,5	12,5	S	Supreme
14	8	1	spinach_supr_s	1	01/01/2015	Thursday	12:51:37 PM	12,5	12,5	S	Supreme
15	9	0,111111111	classic_dlx_s	1	01/01/2015	Thursday	12:52:01 PM	12	12	S	Classic
16	9	0,111111111	green_garden_s	1	01/01/2015	Thursday	12:52:01 PM	12	12	S	Veggie
17	9	0,111111111	ital_cpcllo_l	1	01/01/2015	Thursday	12:52:01 PM	20,5	20,5	L	Classic
18	9	0,111111111	ital_supr_l	1	01/01/2015	Thursday	12:52:01 PM	20,75	20,75	L	Supreme
19	9	0,111111111	ital_supr_s	1	01/01/2015	Thursday	12:52:01 PM	12,5	12,5	S	Supreme
20	9	0,111111111	mexicana_s	1	01/01/2015	Thursday	12:52:01 PM	12	12	S	Veggie
21	9	0,111111111	spicy_ital_l	1	01/01/2015	Thursday	12:52:01 PM	20,75	20,75	L	Supreme
22	9	0,111111111	spin_pesto_l	1	01/01/2015	Thursday	12:52:01 PM	20,75	20,75	L	Veggie
23	9	0,111111111	veggie_veg_s	1	01/01/2015	Thursday	12:52:01 PM	12	12	S	Veggie
24	10	0,5	mexicana_l	1	01/01/2015	Thursday	1:00:15 PM	20,25	20,25	L	Veggie
25	10	0,5	southw_ckn_l	1	01/01/2015	Thursday	1:00:15 PM	20,75	20,75	L	Chicken
26	11	0,25	bbq_ckn_l	1	01/01/2015	Thursday	1:02:59 PM	20,75	20,75	L	Chicken

- **Missing Data Handling:** Checked for missing values using **COUNTBLANK**
- **Feature Engineering:**
 - Created **total_orders** column by using **1/COUNTIF(B:B; [@[order_id]])** to count total orders without duplicates value.
- **Database Integration:** Import file Excel to SQL Server and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions):

We performed structured analysis in SQL Server to answer key business questions:

A. KPI:

1. Total Revenue:

	Total_revenue
1	881654.30

2. Average Order Value:

	Avg_order_value
1	41.295283

3. Total Pizzas Sold:

	Total_pizza_sold
1	49574

4. **Total Orders:**

	Total_orders
1	21350

5. **Average Pizzas Per Order:**

	Avg_pizza_per_order
1	2.32

B. Daily Trend for Total Orders:

	order_days	Total_orders
1	Saturday	3158
2	Wednesday	3024
3	Monday	2794
4	Sunday	2624
5	Friday	3538
6	Thursday	3239
7	Tuesday	2973

C. Hourly Trend for Orders:

	orders_hours	Total_orders
1	9	1
2	10	8
3	11	1231
4	12	2520
5	13	2455
6	14	1472
7	15	1468
8	16	1920
9	17	2336
10	18	2399
11	19	2009
12	20	1642
13	21	1198
14	22	663
15	23	28

D. % of Sales by Pizza Category:

	pizza_category	Total_sales	percent_sales
1	Veggie	193690.45	21.97
2	Chicken	195919.50	22.22
3	Supreme	208197.00	23.61
4	Classic	283847.35	32.19

E. % of Sales by Pizza Size:

	pizza_size	Total_sales	percent_sale
1	L	375318.70	42.57
2	M	249382.25	28.29
3	S	241870.75	27.43
4	XL	14076.00	1.60
5	XXL	1006.60	0.11

F. Total Pizzas Sold by Pizza Category:

	pizza_category	total_pizza_sold
1	Chicken	11050
2	Veggie	11649
3	Supreme	11987
4	Classic	14888

G. Top 5 Best Sellers by Total Pizzas Sold:

	pizza_name	total_pizza_sold
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

H. Bottom 5 Best Sellers by Total Pizzas Sold:

	pizza_name	total_pizza_sold
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Supreme Pizza	950
5	The Soppressata Pizza	961

5. Dashboard in Excel:

Finally, we built an interactive dashboard in Excel to present insights visually.



6. Business Recommendations:

- **Staffing & Operations during Peak Hours:** Increase kitchen and delivery staff during peak windows (12 PM–1 PM and 5 PM–7 PM) and peak days (Friday/Saturday) to speed up order fulfillment.
- **Off-Peak Demand Generation:** Launch "Happy Hour" promotions (2 PM–4 PM) and weekday deals (Monday/Sunday) to drive traffic during low-demand periods.
- **Size Streamlining & Upselling:** Focus upselling campaigns on Size L (42.57% of sales) and consider removing rarely ordered sizes (XL and XXL, under 2% combined) to cut packaging and inventory costs.
- **Menu Rationalization:** Feature top performers (Classic Deluxe, Barbecue Chicken, Hawaiian) prominently in combos, while phasing out or reformulating low-volume items like The Brie Carre (490 orders) to reduce ingredient waste.
- **Increase Average Order Value (AOV):** Bundle pizzas with sides and drinks or offer free delivery on orders above \$45–\$50 to lift the current \$41.30 AOV.